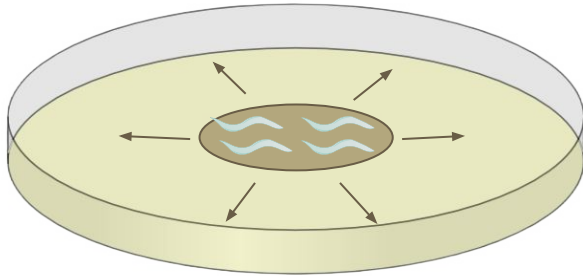

Presenting Figure 3

— Sumaia, Zephyr, Jessica, Tazeem —

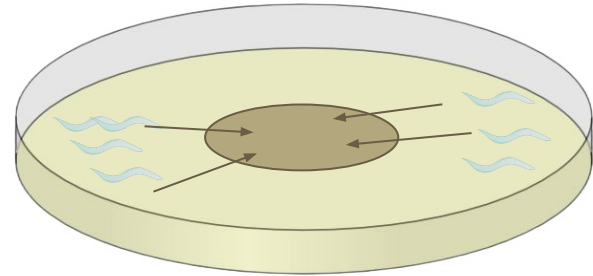
Experimental Set-up

Experimental Design 1- Food Leaving *Figures 3A-3D*



- *C. elegans* start at *e. coli* (food)

Experimental Design 2- Food Reaching *Figures 3E-3H*



- *C. elegans* start 3cm away from *e. coli* (food)

Males have a lower preference for food and a higher preference for exploration than Hermaphrodites

Questions addressed in Figure 3

1. Does *odr-10* controls sex differences in food leaving?
2. Does AWA sex identity controls food leaving?
3. Does misexpression of *odr-10* in AWA alter sex-specific preferences for food?

Questions addressed in Figure 3

1. Does *odr-10* controls sex differences in food leaving?
2. Does *AWA* sex identity controls food leaving?
3. Does misexpression of *odr-10* in *AWA* alter sex-specific preferences for food?

Figure 3A: Does *odr-10* control the difference in food-preference observed between the sexes?

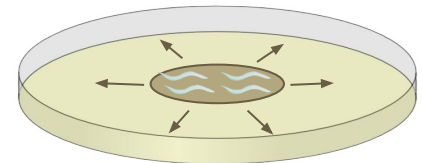
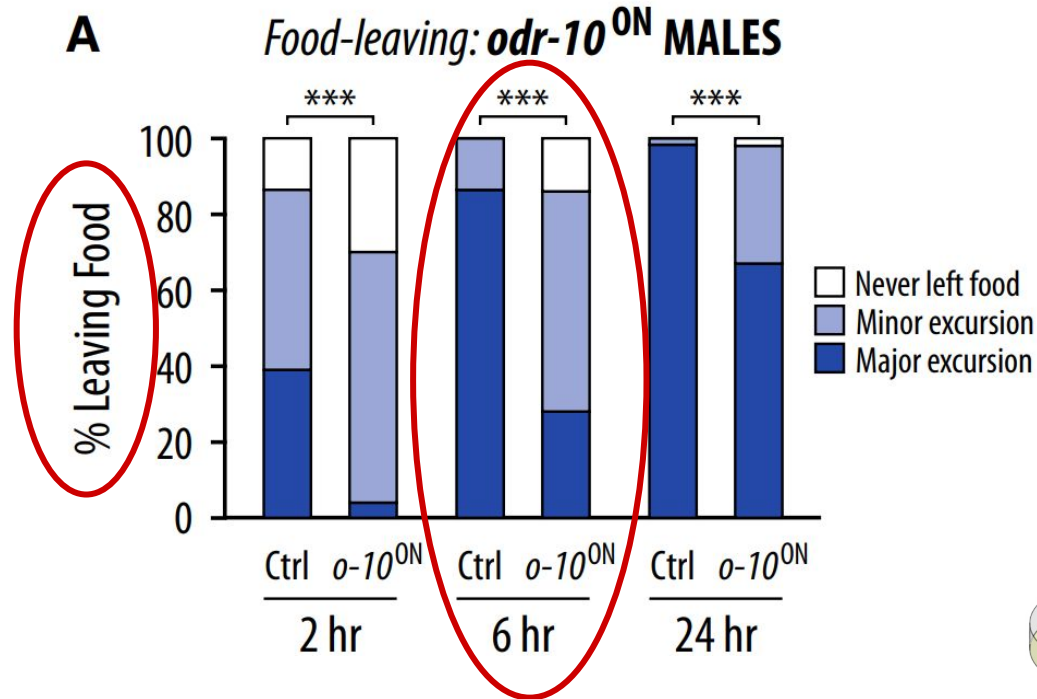


Figure 3E: Does odr-10 control the difference in food-preference observed between the sexes?

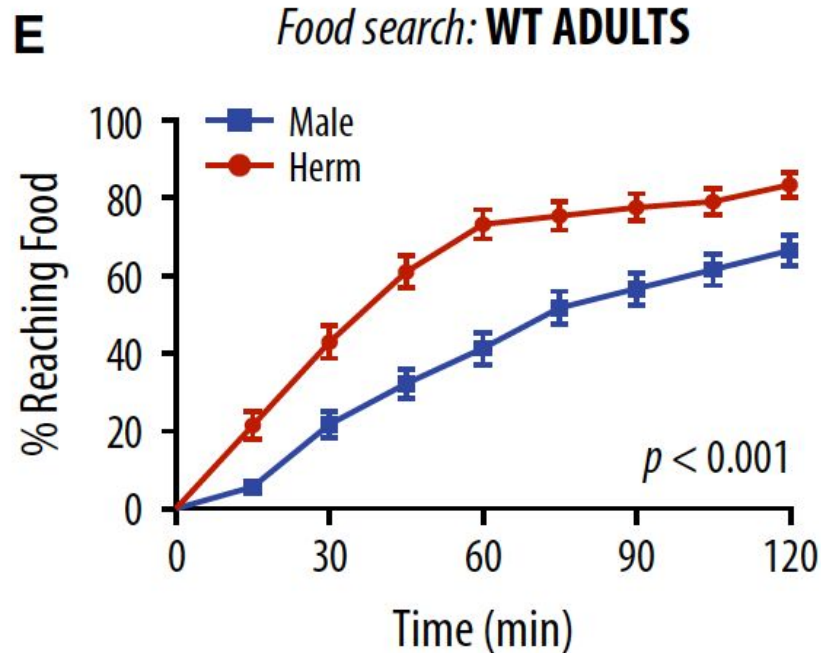
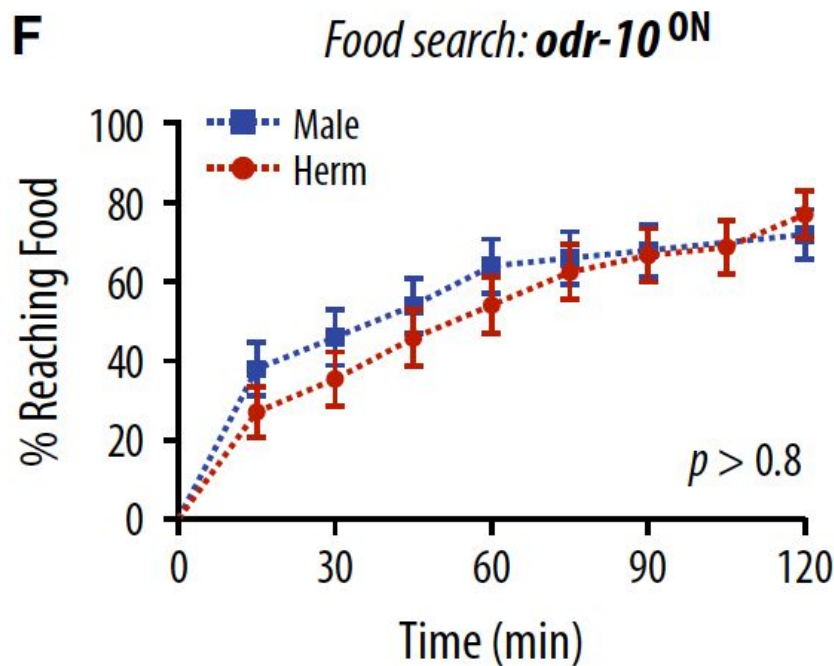


Figure 3F: Does *odr-10* control the difference in food-preference observed between the sexes?



Questions addressed in Figure 3

1. Does *odr-10* controls sex differences in food leaving?
- 2. Does AWA sex identity controls food leaving?**
3. Does misexpression of *odr-10* in AWA alter sex-specific preferences for food?

Figure 3B: Does the AWA sex identity control food leaving?

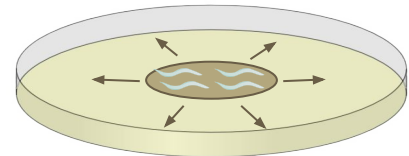
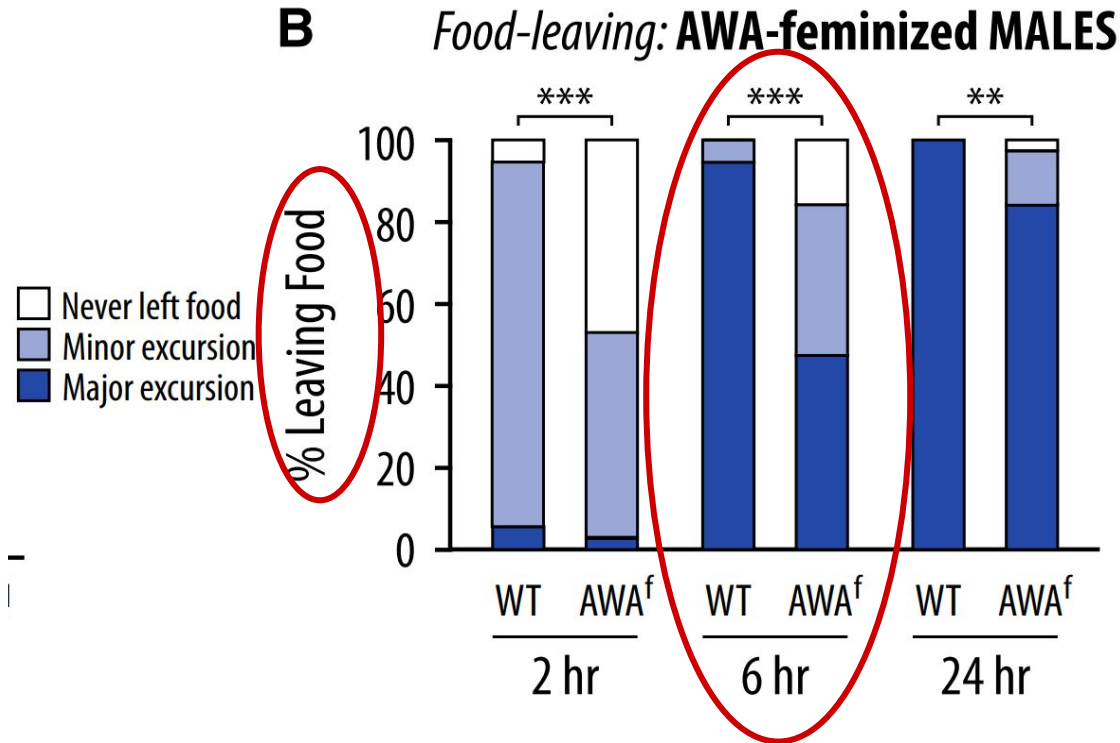
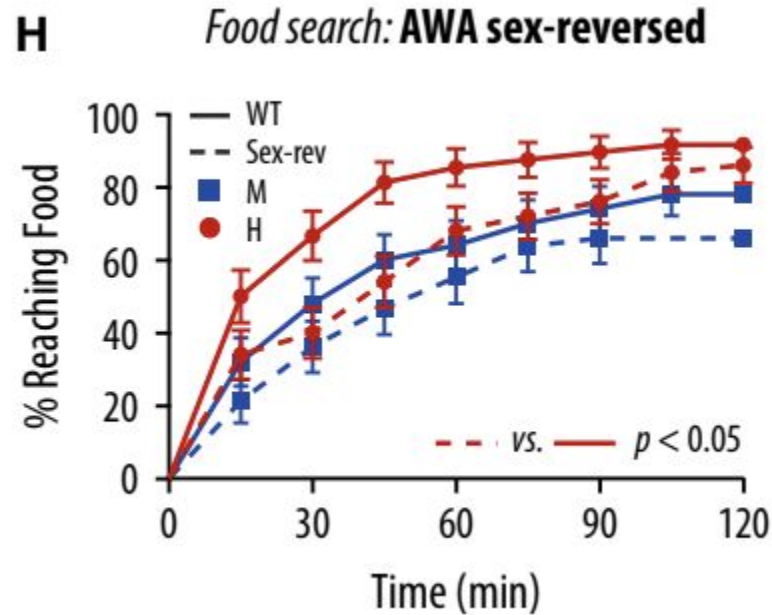


Figure 3H: Does the AWA sex identity control food leaving?



Questions addressed in Figure 3

1. Does *odr-10* controls sex differences in food leaving?
2. Does *AWA* sex identity controls food leaving?
3. Does misexpression of *odr-10* in *AWA* alter sex-specific preferences for food?

Figure 3C: Does misexpression of *odr-10* in AWA alter sex-specific preferences for food?

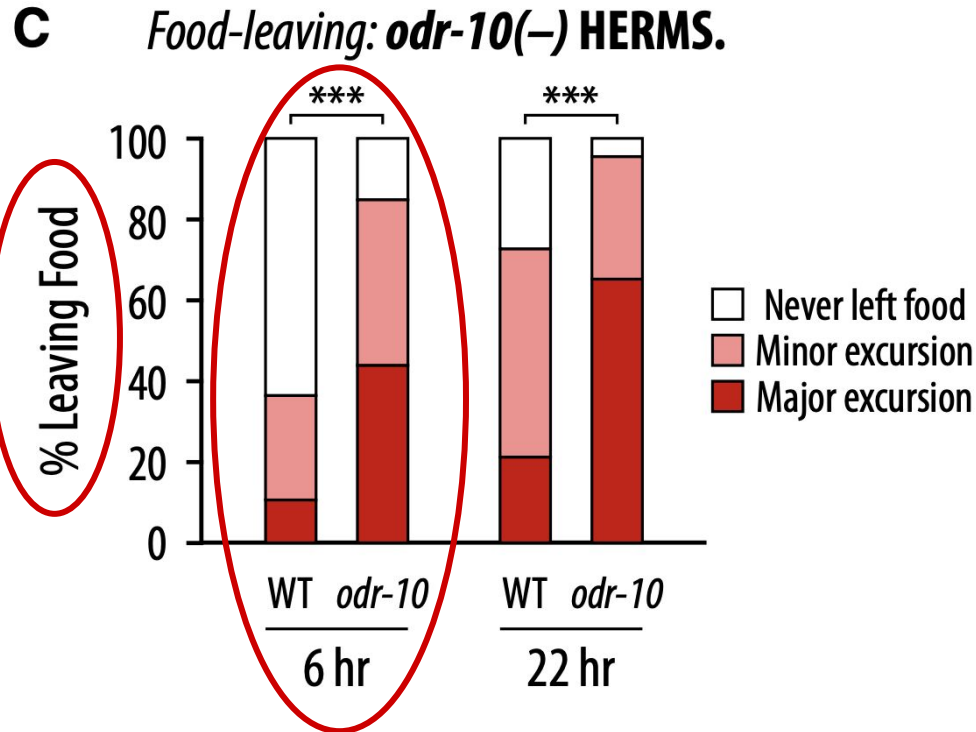


Figure 3D: Does misexpression of *odr-10* in AWA alter sex-specific preferences for food?

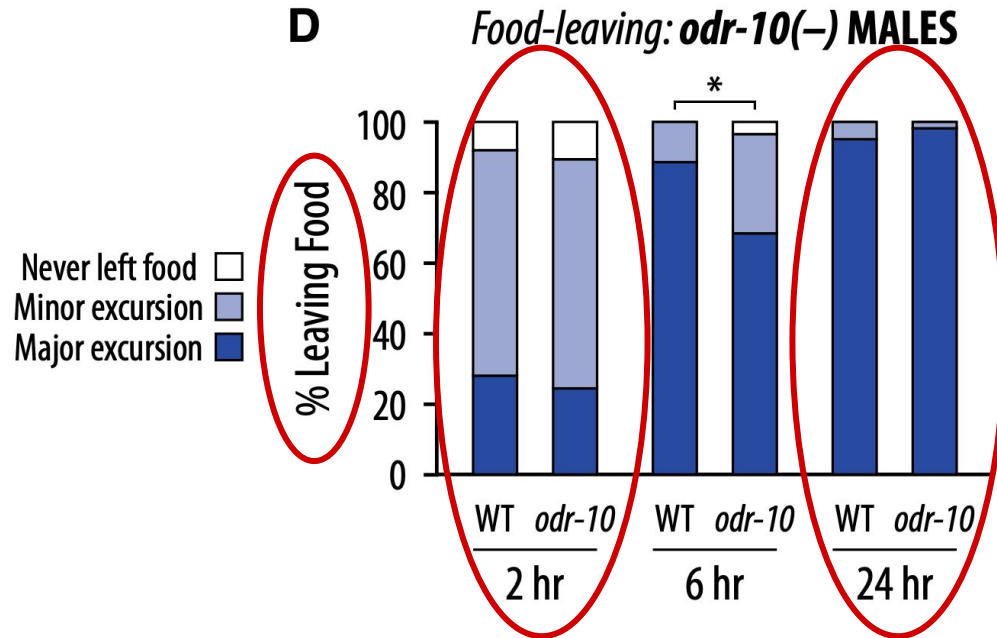
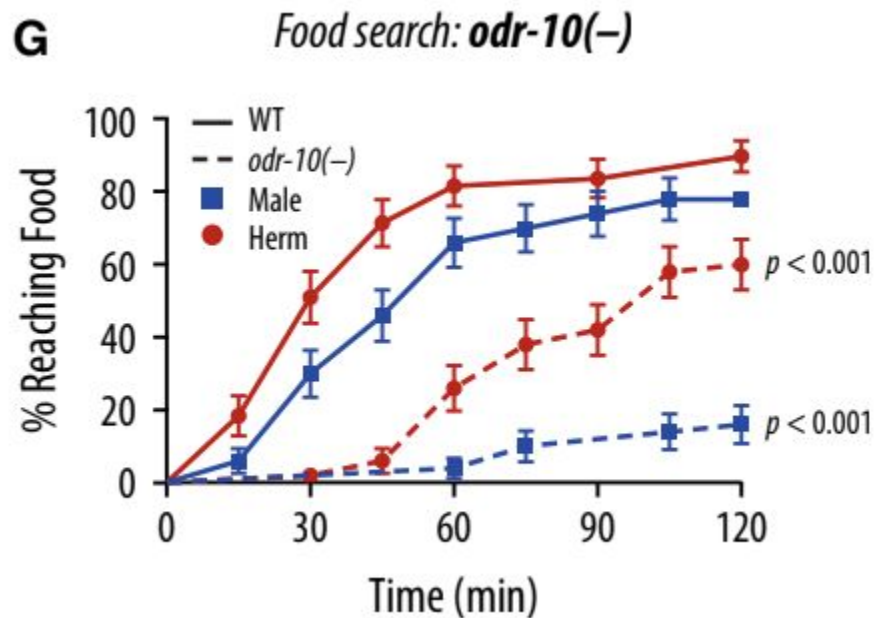
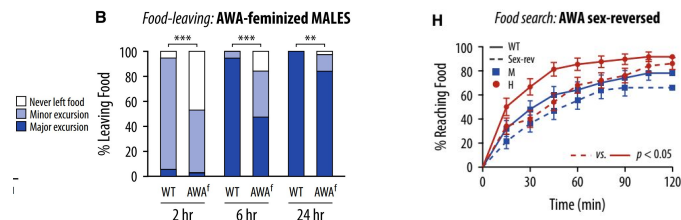


Figure 3G



Conclusions

1. Does *odr-10* controls sex differences in food leaving?
2. Does AWA sex identity controls food leaving?



3. Does misexpression of *odr-10* in AWA alter sex-specific preferences for food?

What is the main question the paper is asking?

Which biological factors, like sex and age, contribute to *c. elegans*' decision whether to prioritize finding food or finding a mate, and how does this process occur? (ZW)

how do different characteristics of the *c.elegans* affect their attraction to food versus mating? (jg)

What causes male *c. elegans* to be less attracted to eating and more attracted to exploration than hermaphrodites? Is it caused by changes in brain/biological signals? (sj)

Why are male *C. elegans* worms more interested in exploring their surroundings and less interested in eating compared to hermaphrodite worms? Is this difference due to changes in certain signals in their brains that affect their behavior? (Tazeem)

What do the authors conclude?

There are sex differences in attraction to diacetyl, an olfactory stimulus that represents a food signal. The cell-autonomous regulation of diacetyl receptor odr-10 influences behavioral differences. Therefore, odr-10 influences adult male c.elegans' behavioral prioritization (exploration or feeding). (sj)

A given c. elegans worm's sex, age, and hunger level determine whether it prioritizes finding food or finding a mate. This process takes place because c. elegans automatically change the extent to which a gene called odr-10, which allows them to smell food, is activated, in response to their environment. (zw)

The chemoreceptor odr-10 plays an important role in detecting food, and the higher the levels of odr-10, the more likely the individual is to search for food. At a juvenile stage, both sexes have the same levels of odr-10 and prioritize food over anything else, but as they get older, the males express less odr-10, meaning they are more likely to prioritize mating over feeding. (jg)

The loss of odr-10 function significantly compromises food-search behavior in both sexes without more generally disrupting sensory behavior. (Tazeem) (note: oversimplified)

3A

Question: How does odr-10 level affect males attraction to food vs exploring their world?

Method: Compared male worms that had wild-type levels of odr-10 and higher levels of odr-10 and looked at percent of time on food compared to exploring.

Conclusion: Males with higher levels of odr-10 spent a higher percentage of time on food and a lower percentage of time exploring than males with wild-type levels. They were more like the hermaphrodites.
